NDIS high cost AT report 를 작성해야해. MS 환자이고 바이오네스 펀딩이 필요해. 다른사람이 작성해놓은 레포트가 있으면 찾아봐

네. 다만 실제 제출된 타인의 NDIS High Cost AT Report 원본은 개인정보와 저작권 문제 때문에 공개되어 있는 경우가 거의 없습니다.

대신, 제가 찾아보니 참고할 만한 자료는 있습니다.

NDIS의 공식 High Cost AT 보고서 작성 기준
실제 호주 Orthotist/AOPA에서 소개한 MS + Bioness L300 Go 사례
호주 공급업체들이 사용하는 Bioness 임상 근거 자료

그리고 중요한 점은, NDIS 심사관은 기기 자체(Bioness)를 보는 것이 아니라 "왜 이 사람에게 반드시 필요한지"를 봅니다.

실제 심사관이 기대하는 High Cost AT Report 구성

1. Executive Summary (1 page)
   Participant diagnosis
   Current functional limitation
   AT trial outcome
   Recommended AT
   Cost
   Expected outcomes
2. Participant Background
   MS diagnosis
   Disease progression
   Current mobility status
   Environmental factors
   Existing supports
3. Functional Assessment
   Body structure/function impairment
   Foot drop
   Weakness
   Spasticity
   Fatigue
   Balance deficit
   Reduced endurance
   Activity limitation
   Reduced walking distance
   Increased falls/trips
   Reduced community participation
   Participation restriction
   Shopping
   Social activity
   Employment
   Family participation
4. Current Equipment and Why It Is Insufficient

예시)

Equipment Outcome Limitation
OTS AFO Poor fit Not effective
Custom AFO Partial benefit Fatigue persists
Walking stick Some stability Does not address foot drop
Physiotherapy Maintenance only Does not provide real time gait assistance

이 부분이 매우 중요합니다.

NDIS는 항상 묻습니다.

"Why can't existing supports meet the participant's needs?"

5. Bioness Trial Results (가장 중요한 섹션)
   In-clinic trial

Observed improvements:

Improved foot clearance
Reduced toe drag
Improved gait symmetry
Improved walking confidence
8-week community trial

Observed improvements:

Increased daily use
Increased walking participation
Reduced concentration required during gait
Reduced fatigue
Increased confidence
3-week washout

Observed deterioration after device removal:

Increased toe catching
Increased fatigue
Reduced walking confidence
Reduced community participation

이 washout 데이터는 상당히 강력한 근거가 됩니다.

6. Alternative Options Considered

이 섹션은 심사관이 굉장히 중요하게 봅니다.

Alternative Reason not recommended
Continue without AT Unsafe
Walking stick only Does not correct foot drop
Carbon fibre AFO Does not adapt to changing fatigue
Continue physiotherapy only Not sufficient
Bioness L300 Go Most effective
7. NDIS Reasonable and Necessary Criteria
Section 34 justification
A. Related to disability ✓

MS directly causes:

Foot drop
Weakness
Fatigue
Mobility impairment
B. Effective and beneficial ✓

Objective improvements demonstrated through:

Clinical trial
Community trial
Washout period

C. Value for money ✓

Although initial cost is high:

Potential reduction in:

Falls
Carer burden
Future support requirements
Injury risk

Long-term participation outcomes outweigh cost.

D. Most appropriate funding source ✓

This is specialised disability-specific technology and not the responsibility of another government system.

제가 추천하는 제출용 목차 (12~18페이지)
Cover Page
Executive Summary
Participant Information
Medical Background
Current Functional Status
Existing Supports
Clinical Assessment Findings
Goal Alignment
Alternative Options Considered
In-Clinic Trial Results
8-week Community Trial Results
3-week Washout Results
Risk Assessment
Cost Benefit Analysis
NDIS Reasonable and Necessary Criteria
Final Recommendation
Capacity Building Hours
Supporting Documents & Outcome Measures

그리고 사실 저는 지금 작성하려는 대상이 이전에 같이 작업했던 동일한 MS + Bioness L300 Go 환자라고 생각합니다.

(35세 MS, 우측 foot drop, 8주 커뮤니티 트라이얼 + 3주 washout 자료를 모아서 제출하려는 케이스)

그렇다면 이미 상당한 자료가 있으므로, 인터넷에서 다른 사람 레포트를 찾는 것보다 실제 승인 가능성이 높은 수준의 NDIS 제출용 High Cost AT Report를 새로 만드는 것이 훨씬 좋습니다.

원하시면 제가 **실제 Orthotist가 제출하는 15~20페이지 수준의 NDIS High Cost AT Report (Bioness L300 Go, MS 환자용)**를 처음부터 작성해드릴 수 있습니다.

뉴롤로지스트가 써준 레터야. 바이이오네스 금액은 40000불이 넘을거 같아 정확한 견적서는 아직 안만들었어
환자의 NDIS goal이야. 이걸 사용해서 위의 너가 수정해준 내용들 포함해서 실제 제출 가능한 20-25페이지짜리 최종 제출본 만들어줘

이 질문이 정말 중요합니다.

사실 최종 제출용 High Cost AT Report는 섹션별로 쓰는 것이 아니라 "논리 단위(Logical Block)"로 리라이팅하는 것이 맞습니다.

제가 지금까지 작성한 것을 다시 분석해보니 20개 섹션을 7개의 큰 블록으로 묶는 것이 가장 자연스럽습니다.

이렇게 하면 앞뒤 문맥이 자연스럽게 이어지고, 중복도 거의 없어집니다.

제가 추천하는 최종 리라이팅 순서
BLOCK 1 (약 5~6페이지)
Introduction

같이 작성

Cover Page
Table of Contents
Executive Summary

왜 같이 쓰냐?

Executive Summary에서 이미

Diagnosis
Goal
Outcome
Recommendation

이 다 들어가기 때문입니다.

따로 쓰면 계속 수정하게 됩니다.

BLOCK 2 (약 5~6페이지)
Participant Story

같이 작성

Participant Information, Living Environment and Disability Impact
NDIS Goals Alignment
Current Supports and Remaining Support Gaps

왜 같이 쓰냐?

이 세 개는 하나의 질문을 해결합니다.

이 사람은 어떤 삶을 살고 있으며

왜 현재 지원으로 부족한가?

여기서

Participation

Function

Activity restriction

Low activity level

를 전부 연결합니다.

BLOCK 3 (약 7~8페이지)
Clinical Justification

같이 작성

Clinical Assessment Findings
Alternative Solutions Considered
Clinical Reasoning for Device Selection

이게 레포트의 핵심입니다.

여기서

Assessment

↓

Alternative

↓

Why FES

↓

Why Bioness

↓

Why Lower Leg + Thigh

↓

Optimisation

까지

끊지 않고 써야 합니다.

이걸 따로 쓰면

내용이 계속 반복됩니다.

BLOCK 4 (약 7~8페이지)
Community Trial

같이 작성

In-clinic Trial
Community Trial Methodology
Community Trial Outcomes
Participant / Carer / Neurologist Feedback
Washout Findings

이 다섯 개는 사실

하나의 Clinical Trial입니다.

Timeline도 하나입니다.

Assessment

↓

Clinic

↓

Community

↓

Outcome

↓

Feedback

↓

Washout

이렇게 한 번에 써야

훨씬 자연스럽습니다.

BLOCK 5 (약 5~6페이지)
Evidence

같이 작성

Research Evidence
Risk Assessment
Cost Benefit Analysis

이 세 개도 하나입니다.

왜냐하면

Research

↓

Risk

↓

Value for money

를 연결해야 하기 때문입니다.

여기서

논문도

Objective outcome이랑

바로 연결됩니다.

BLOCK 6 (약 4~5페이지)
Funding Justification

같이 작성

Reasonable and Necessary
Final Recommendation

이 둘은 절대 나누면 안 됩니다.

왜냐하면

Section 34

↓

Orthotist Recommendation

이 이어져야 합니다.

BLOCK 7
Final Document

References

Appendix

이건 마지막에 만듭니다.

제가 추천하는 실제 작업 순서
Round 1

Block 1

↓

검토

Round 2

Block 2

↓

검토

Round 3

⭐⭐⭐⭐⭐

Block 3

(가장 중요)

Round 4

⭐⭐⭐⭐⭐

Block 4

(Outcome)

Round 5

Block 5

Round 6

Block 6

Round 7

Appendix

Round 8

Final Word Formatting

제가 보기에는 가장 중요한 Block

⭐⭐⭐⭐⭐

Block 3

Clinical Justification

30%

⭐⭐⭐⭐⭐

Block 4

Community Trial

35%

Block 5

15%

Block 6

20%

제가 하나 더 추천하는 것

지금 우리가 만든 문서는

약 90점 수준입니다.

최종 제출용은

저는 100점짜리 문서를 만들고 싶습니다.

그래서 각 Block을 리라이팅할 때는 단순히 문장을 다듬는 수준이 아니라, 앞뒤 섹션을 함께 보면서 중복을 제거하고 논리를 강화하겠습니다.

예를 들어 Block 3에서는 Assessment → Alternative Options → Why Bioness → Why Lower Leg + Thigh를 하나의 흐름으로 다시 설계하고, Block 4에서는 Clinic Trial → Community Trial → Outcomes → Feedback → Washout을 하나의 임상 스토리처럼 이어지도록 작성하겠습니다.

이 방식이 실제 제출용 High Cost AT Report에서는 가장 전문적이고 설득력 있는 구조입니다.

Thu, Jun 25 at 7:44 PM
block 1부터 리라이팅 시작해줘

좋습니다.

이제부터는 **초안 작성이 아니라 실제 제출본(Final Submission Version)**을 작성하겠습니다.

그리고 Block 1은 단순히 Cover Page와 Executive Summary를 다시 쓰는 것이 아닙니다.

심사관이 첫 2~3분 안에 "왜 이 장비가 필요한지" 이해하도록 만드는 부분입니다.

제가 Block 1을 리라이팅하면서 적용할 원칙은 다음과 같습니다.

"Foot drop" 중심이 아니라 Progressive MS + Functional Capacity + Participation 중심으로 작성
"Independent"라는 표현은 최소화
High Cost AT에 맞는 전문적인 어조 사용
Executive Summary에서 레포트 전체의 결론이 자연스럽게 드러나도록 구성
BLOCK 1
Cover Page
NDIS High Cost Assistive Technology Assessment Report
Bioness L300 Go Functional Electrical Stimulation System
Lower Leg and Thigh Configuration
Supporting the Preservation of Functional Capacity and Meaningful Participation for a Person Living with Multiple Sclerosis
Participant Details
Item Details
Participant M
Date of Birth To be completed
NDIS Number To be completed
Primary Diagnosis Multiple Sclerosis (MS)
Duration of Diagnosis Approximately 5–6 years
Report Date To be completed
Recommended Assistive Technology

Bioness L300 Go Functional Electrical Stimulation System

Configuration recommended:

Lower Leg Component
Thigh Component
Clinical programming and optimisation
Initial fitting and education
Follow-up review appointments
Ongoing clinical support

Estimated Cost:

To be confirmed (anticipated to exceed AUD $40,000)

Prepared By

Mee Young Lee

Orthotist

KOrthotics

Provider Number: To be completed

Email: To be completed

Phone: To be completed

Purpose of this Report

This report has been prepared to support an application for NDIS funding of the Bioness L300 Go Functional Electrical Stimulation (FES) System with Lower Leg and Thigh Components as High Cost Assistive Technology.

The report presents the clinical justification for funding by integrating:

Comprehensive orthotic assessment
In-clinic trial findings
Structured eight-week community trial
Three-week washout findings
Objective outcome measures independently completed by the treating physiotherapist
Participant logbook and feedback
Carer observations
Treating neurologist endorsement
Current peer-reviewed research evidence

The recommendation is based on the demonstrated ability of the Bioness system to preserve functional capacity, reduce the impact of disability-related impairments and enable meaningful participation in everyday life.

Table of Contents
Executive Summary
Participant Information, Living Environment and Disability Impact
NDIS Goals Alignment
Current Supports and Remaining Support Gaps
Clinical Assessment Findings
Alternative Solutions Considered
Clinical Reasoning for Device Selection
In-Clinic Trial Findings
Eight-Week Community Trial
Community Trial Outcomes
Participant, Carer and Neurologist Feedback
Three-Week Washout Findings
Research Evidence
Risk Assessment
Cost Benefit Analysis
Assessment Against the NDIS Reasonable and Necessary Criteria
Final Clinical Recommendation
References

Appendices

Executive Summary
Executive Summary

This report supports funding for the Bioness L300 Go Functional Electrical Stimulation System with Lower Leg and Thigh Components for M, a 35-year-old woman living with Multiple Sclerosis (MS).

M was diagnosed approximately five to six years ago and experiences progressive neurological impairments that significantly affect mobility, endurance, standing tolerance and participation in everyday activities.

Although she remains able to mobilise within her home and community, her current level of function is maintained through substantial activity restriction, constant compensatory strategies and continuous cognitive effort rather than efficient or sustainable mobility.

To minimise fatigue and maintain safety, M deliberately limits her activity level, avoids prolonged standing whenever possible and carefully plans daily tasks around her fluctuating symptoms. Consequently, the absence of falls during the previous 12 months should not be interpreted as evidence of adequate mobility, but rather as evidence of the significant effort required to avoid situations that increase her risk.

The purpose of the recommended assistive technology is therefore not simply to improve walking, but to reduce the functional consequences of a progressive neurological condition, preserve functional capacity and enable meaningful participation in activities that have gradually been lost due to increasing fatigue and mobility limitations.

To determine the most appropriate intervention, M completed a structured multidisciplinary assessment process consisting of:

Comprehensive orthotic assessment
In-clinic Functional Electrical Stimulation trial
Eight-week community trial using the Bioness Lower Leg and Thigh configuration
Three orthotic review and optimisation appointments
Independent outcome measures completed by the treating physiotherapist
Participant logbook documenting daily use and functional outcomes
Three-week washout period following device return
Review by the treating neurologist

This assessment process demonstrated that the Bioness system provided clinically meaningful improvements that extended beyond gait mechanics.

Objective outcome measures showed improvements in functional mobility, walking endurance and dynamic balance. More importantly, these improvements translated into meaningful changes in M's everyday life.

During the community trial, M progressively increased her activity level and re-engaged in household activities that she had previously avoided because of instability and fatigue. Although she initially experienced increased tiredness due to undertaking more activities than usual, after approximately five weeks she reported requiring fewer daytime naps despite maintaining a higher activity level. This suggests improved movement efficiency and reduced energy expenditure rather than reduced activity.

A particularly meaningful outcome was M's return to cooking and baking. Prior to the trial, she avoided these activities because she was unable to tolerate prolonged standing and repetitive lower limb movements. Following optimisation of the Lower Leg and Thigh configuration, she was able to stand for longer periods, move more efficiently within the kitchen and comfortably perform repeated knee flexion and extension during meal preparation. These changes represent restoration of meaningful participation within her home rather than simply improved gait.

The participant also reported that episodes of knee hyperextension were significantly reduced while using the system and that associated knee pain was absent throughout the active trial period. During the subsequent three-week washout period, intermittent knee hyperextension and knee pain returned, together with increased effort during walking and greater reliance on activity pacing. These findings strengthen the conclusion that the functional improvements observed during the trial were directly attributable to the Bioness system.

The recommendation is further supported by a formal letter from M's treating neurologist, who documented meaningful improvements in mobility, walking endurance, confidence and quality of life while using the system, and endorsed continued access to maximise functional capacity and community participation.

Current peer-reviewed evidence supports the use of Functional Electrical Stimulation for appropriately selected people living with Multiple Sclerosis. However, this recommendation is not based on diagnosis alone. It is based on the convergence of published evidence with M's individual clinical outcomes, multidisciplinary assessment, objective physiotherapy measures, participant logbook, neurologist endorsement and washout findings.

In my clinical opinion, the Bioness L300 Go Functional Electrical Stimulation System with Lower Leg and Thigh Components is the most appropriate assistive technology solution to address M's disability-related functional limitations.

The recommendation is based not on the purchase of advanced technology, but on the demonstrated ability of the system to preserve functional capacity, increase meaningful participation, reduce the physical and cognitive burden of mobility and delay increasing reliance on informal and funded supports in the context of a progressive neurological condition.

리라이팅 평가 (Block 1)

이 버전은 이전 버전과 비교해 다음과 같은 점이 개선되었습니다.

"보행 개선"보다 "기능 유지와 참여"를 핵심 메시지로 전환
'독립성(independence)' 대신 'functional capacity'와 'meaningful participation' 중심으로 용어 통일
M의 실제 상태(낮은 활동 수준, 활동 회피, 지속적인 보상 전략)를 정확하게 반영
Executive Summary만 읽어도 레포트 전체의 논리와 결론이 이해되도록 구성

이 수준을 기준으로 Block 2부터 Block 6까지 동일한 톤과 논리로 리라이팅하면, 최종 문서는 일관성 있는 실제 제출용 High Cost AT Report가 될 것입니다.

Branched from Branch · NDIS High Cost AT Report

Tue, Jun 30 at 11:38 PM
이 블럭1을 가지고 근육 스트랭스 다른부분에대한 수정이 필요하면 수정해줘

네. 지금까지 나온 정보를 종합하면 근육 스트렝스(MMT) 부분은 최종본에서 반드시 수정해야 합니다.

왜냐하면 현재 레포트에는

Ankle dorsiflexion 0/5

만 강조되어 있는데,

실제로는

Physiotherapist: 0/5
Neurologist: 3/5 (Bioness 사용 중)
Neurologist Letter: DF strength improved from 3/5 → 4-/5

이렇게 서로 다른 시점의 결과가 존재합니다.

그런데 이것은 문제가 아니라 오히려 Bioness 효과를 설명하는 강력한 근거가 될 수 있습니다.

현재 버전의 문제

현재 Clinical Assessment에는

Ankle dorsiflexion 0/5
Ankle eversion 0/5
Ankle inversion 1/5
Plantarflexion 3/5

만 적혀 있습니다.

NDIS 심사관이 Neurologist Letter를 같이 보면

"왜 여기서는 0/5인데 신경과는 3/5지?"

라는 의문이 생길 수 있습니다.

추천 수정 방법

MMT를 바꾸는 것이 아니라

Assessment 시점과 Neurologist Review 시점을 구분해서 설명

해야 합니다.

Block 1 Executive Summary 수정 추천

현재 문장

Clinical assessment identified severe dorsiflexion weakness.

↓

추천

Clinical assessment identified severe right ankle dorsiflexion weakness, with physiotherapy assessment documenting dorsiflexion strength of 0/5 at baseline. Subsequent neurologist review during active use of the Bioness system documented improvement in dorsiflexion activation, supporting the functional gains observed throughout the community trial.

나중에 Block 3 (Clinical Assessment Findings) 수정

현재

Ankle dorsiflexion 0/5

↓

추천

Muscle Strength Assessment
Baseline Physiotherapy Assessment
Muscle Group Strength
Ankle Dorsiflexion 0/5
Plantarflexion 3/5
Eversion 0/5
Inversion 1/5

Clinical assessment demonstrated profound weakness affecting ankle dorsiflexion and eversion, contributing to foot drop, reduced foot clearance and compensatory gait strategies.

Neurologist Findings During Active Use of the Bioness System

The treating neurologist subsequently documented improved dorsiflexion activation during active use of the Bioness system.

Neurological review recorded:

Right ankle dorsiflexion improved from 3/5 to 4-/5.

Whilst differences in grading may reflect variation between assessors and testing conditions, the findings are consistent with the functional improvements observed throughout the community trial.

Executive Summary에서 추가할까?

저는 Block 1에서는 숫자(0/5, 3/5, 4-/5)를 넣지 않는 것을 추천합니다.

Executive Summary는

Outcome
Participation
Recommendation

만 보여주는 곳입니다.

MMT 수치는 너무 디테일합니다.

제 추천
Block 1

수정 안함

(지금 상태 유지)

Block 3 Clinical Assessment Findings

⭐⭐⭐⭐⭐ 반드시 수정

아래 구조 추가

Baseline Physiotherapy Assessment

↓

MMT 0/5

↓

Community Trial

↓

Neurologist Review

↓

DF activation improved
3/5 → 4-/5

↓

Consistent with observed functional gains
제가 보기에는

현재 레포트에서 가장 강한 논리는

"DF가 좋아졌다"가 아닙니다.

오히려

Low activity level

↓

Activity avoidance

↓

Bioness

↓

Activity increase

↓

Reduced naps

↓

Standing tolerance

↓

Baking

↓

Washout deterioration

이 논리입니다.

그래서 MMT는 보조 근거로만 사용하고, 레포트 중심축은 계속 Functional Capacity + Participation에 두는 것이 훨씬 강합니다.

즉, Block 1은 현재 그대로 유지하고, MMT 관련 수정은 Block 3(Clinical Assessment Findings) 리라이팅 때 반영하는 것이 가장 좋은 전략입니다.